



AFRL

AIR FORCE RESEARCH LABORATORY INFORMATION DIRECTORATE

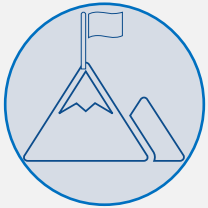
PRESENTATION TO
NAME | COMPANY

NAME
TITLE | AFRL INFORMATION DIRECTORATE



Air Force Research Laboratory

Mission & Vision



AFRL MISSION:

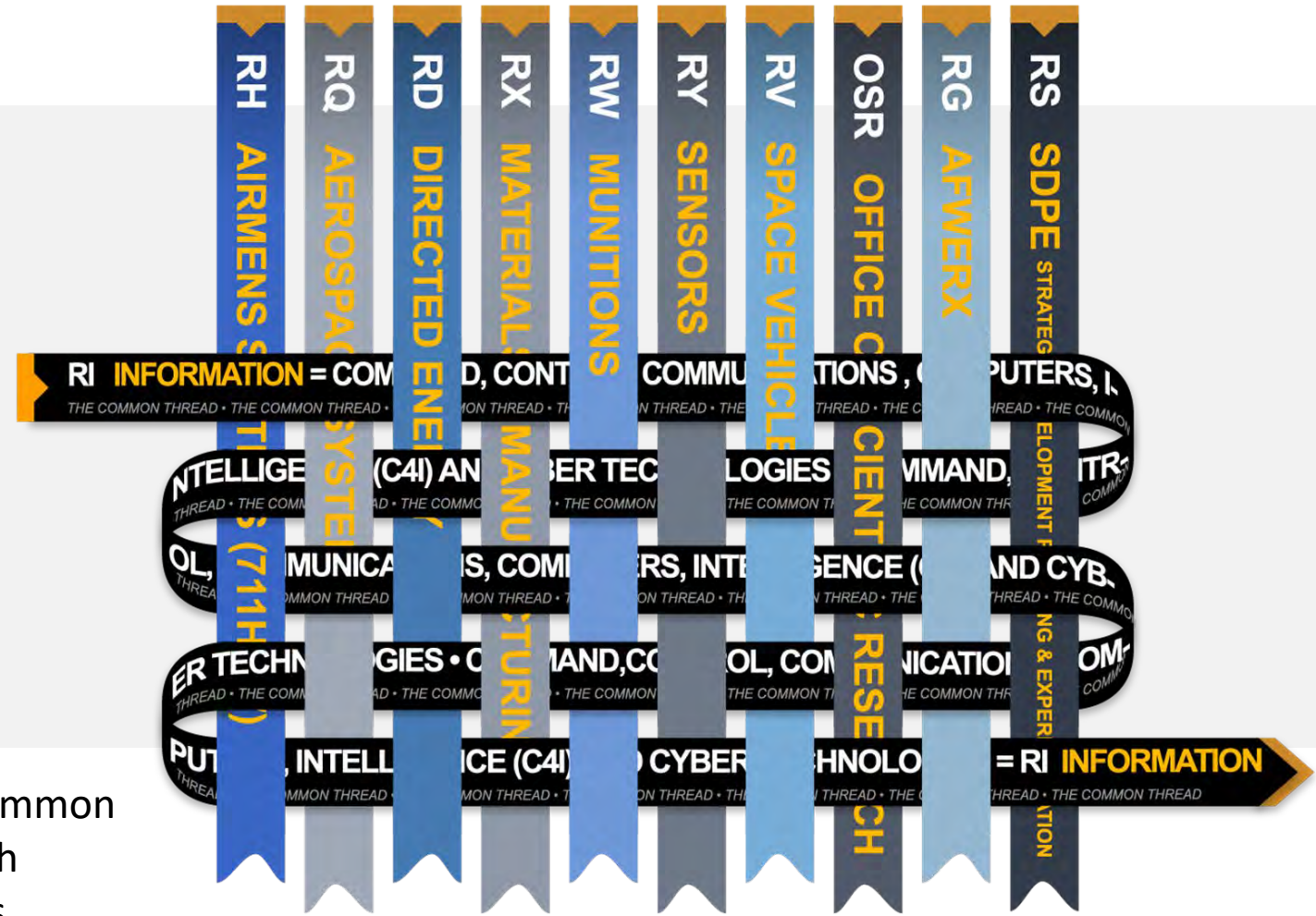
Discover, develop, and deliver high-impact S&T



AFRL VISION:

Win the Future

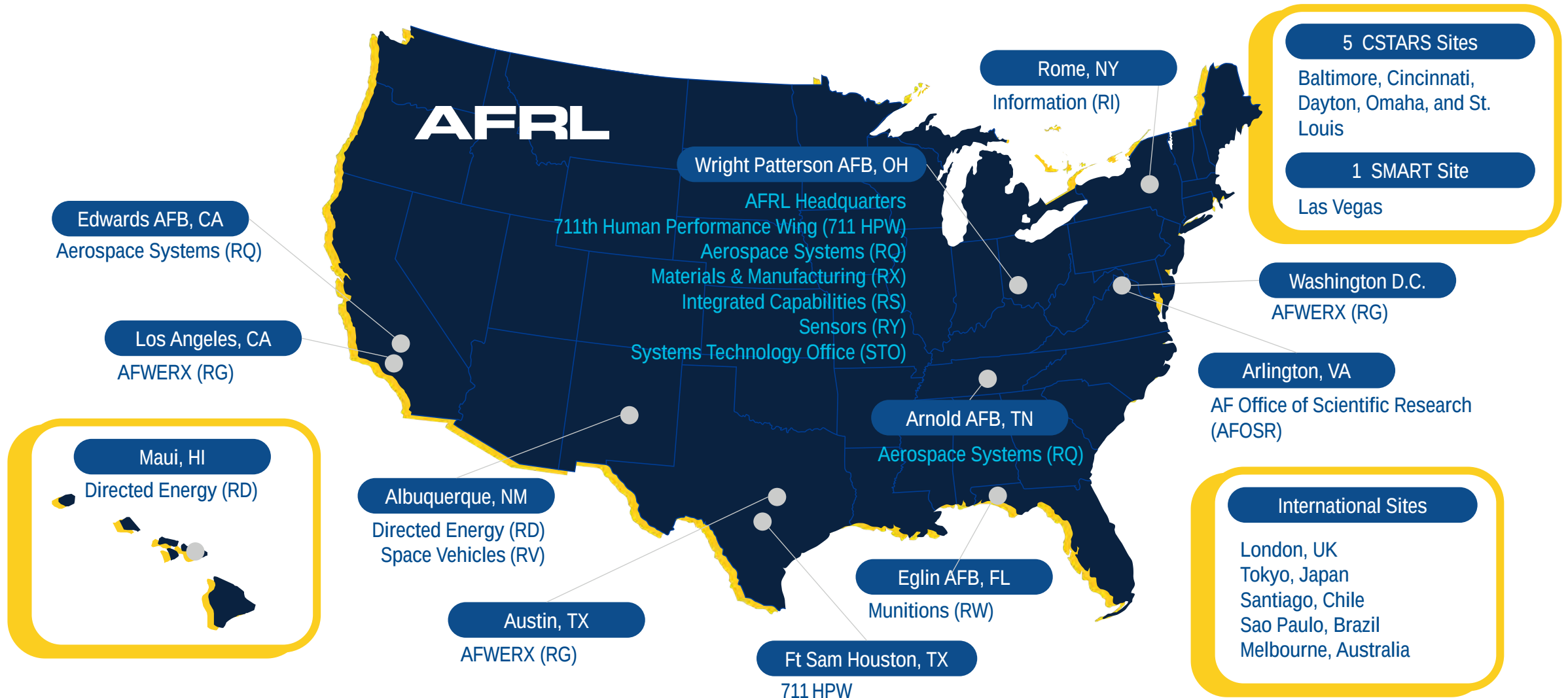
The Information Directorate is the common thread between all Air Force Research Laboratory directorates and locations





Air Force Research Laboratory Locations

*C-STARS: Center for the Sustainment of Trauma and Readiness Skills
*SMART: Sustained Medical and Readiness Trained





Air Force Research Laboratory

Information Directorate Mission & Vision



MISSION:

To discover, demonstrate, and deliver high-impact C5ISRT technologies that enhance capabilities and assure technical superiority for DAF and its partners.



VISION:

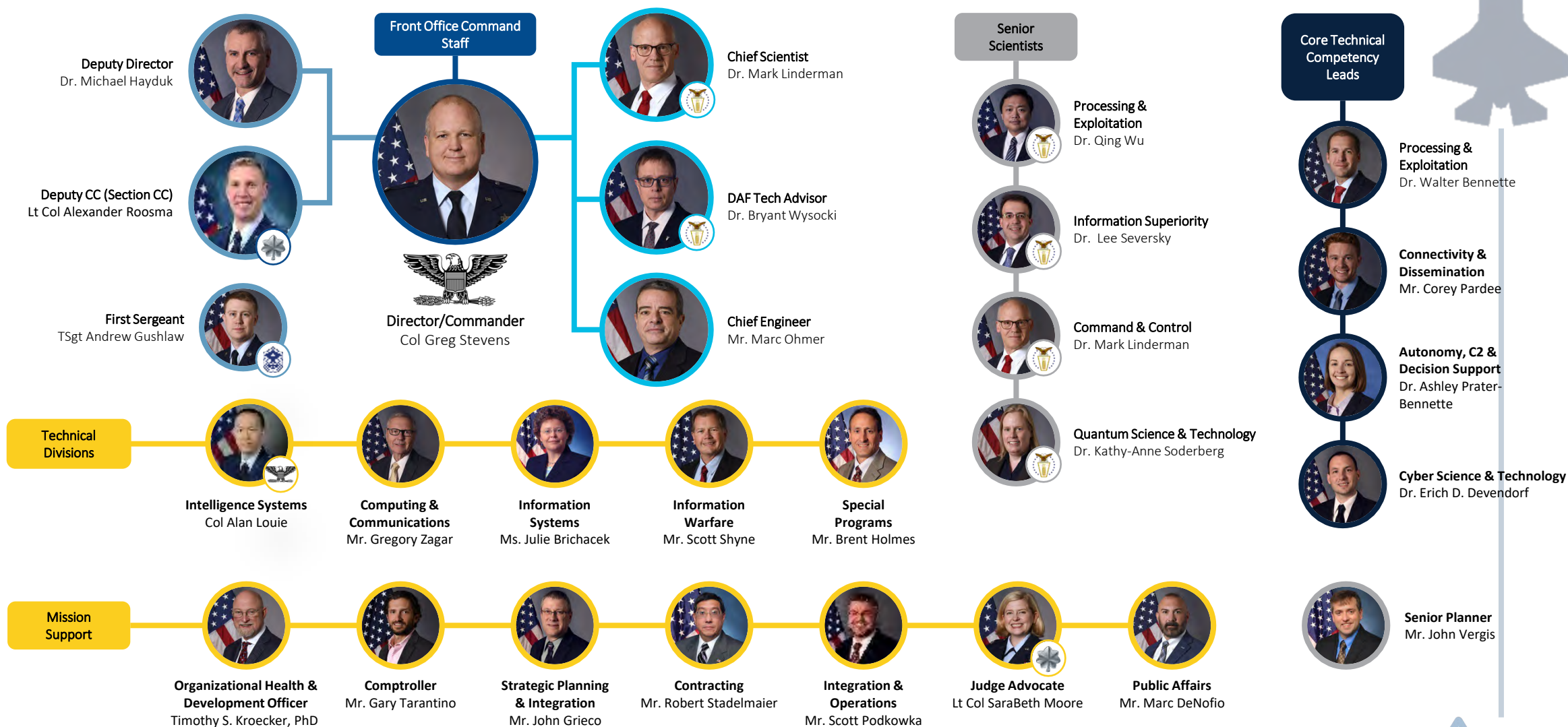
Dominate the information age.



The Information Directorate focus is



Information Directorate Leadership



Historic Moments from the Labs



Minicard Intelligence Data Handling System

Online search is older than you think



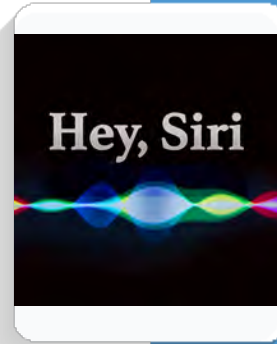
ARPA Network RADC

The early internet and link To Information Computing Service



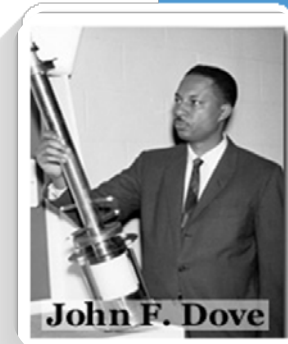
First Communication Satellite Echo 1

Communication signals were transmitted from one location on Earth and bounced off the surface of the satellite to another Earth location



Cognitive Assistant That Learns and Organizes (CALO)

An artificial intelligence project that attempted to integrate numerous AI technologies into a cognitive assistant predating SIRI



John F. Dove Laser Disc Technology Creator

Dove while at RADC, invented and held the patent for laser light technology used in development of the compact disc (commonly known as CD-ROM), video disc and compact disc drives.



Micro-Electro-Mechanical Systems (MEMS)

a miniature machine that has both mechanical and electronic components such as in smartphone accelerometers and gyroscopes for smartphones motion sensing.



A rich heritage of research innovation



Surveillance Radar



PAVE Mover



Airborne Digital Map System



IR Camera for B-52



Moving Target Indicators Experiment



Single Pass AirDrop



Selective Cyber Operations Technology Integration



ECHO-I SATCOM (1st SAT Comm)



Russian to English machine translation



3D Memory



Advanced Planning System



Multi-Level Security



Cyber Situational Awareness



NSDC



Rome Air Development Center
Established 1951 – 1991



Rome Laboratory
Established 1991 – 1997



AF Research Lab Information Directorate
Established 1997 – Present



Intelligence Data Handling Systems



Skylab Tracking



SEM-E Modules For the F-22



Software Programmable Radio (forerunner of JTRS)



CONDOR Supercomputer



DCGS



DARPA's agent for ARPANET



Research Facility Newport & Stockbridge



Track & ID Fusion Algorithms for AWACS



Off-Board Data On J-STARS



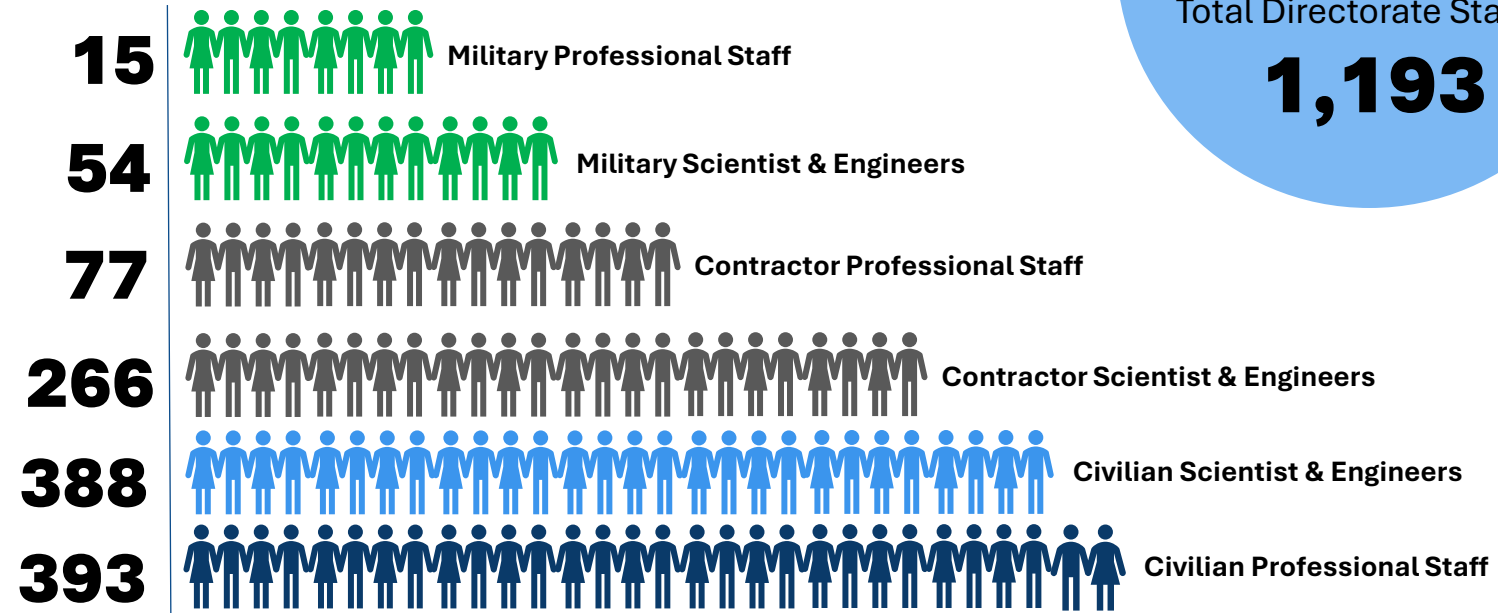
WebTAS



Talent at the Information Directorate

AN EFFECTIVE, EFFICIENT, CROSS-FUNCTIONAL TEAM GOVERNMENT CAREER CLASSIFICATIONS

- Electronics Engineering
- Computer Science
- Engineering (General, Computer, Mechanical, Materials, Civil, Environmental, Industrial)
- Physics
- Mathematics
- Operations Research
- Telecommunications
- Human Resources Management
- Logistics Management
- Public Affairs
- Architecture
- Business Administration
- Financial Management
- Contracting
- Patent Attorney & Legal Services
- Police

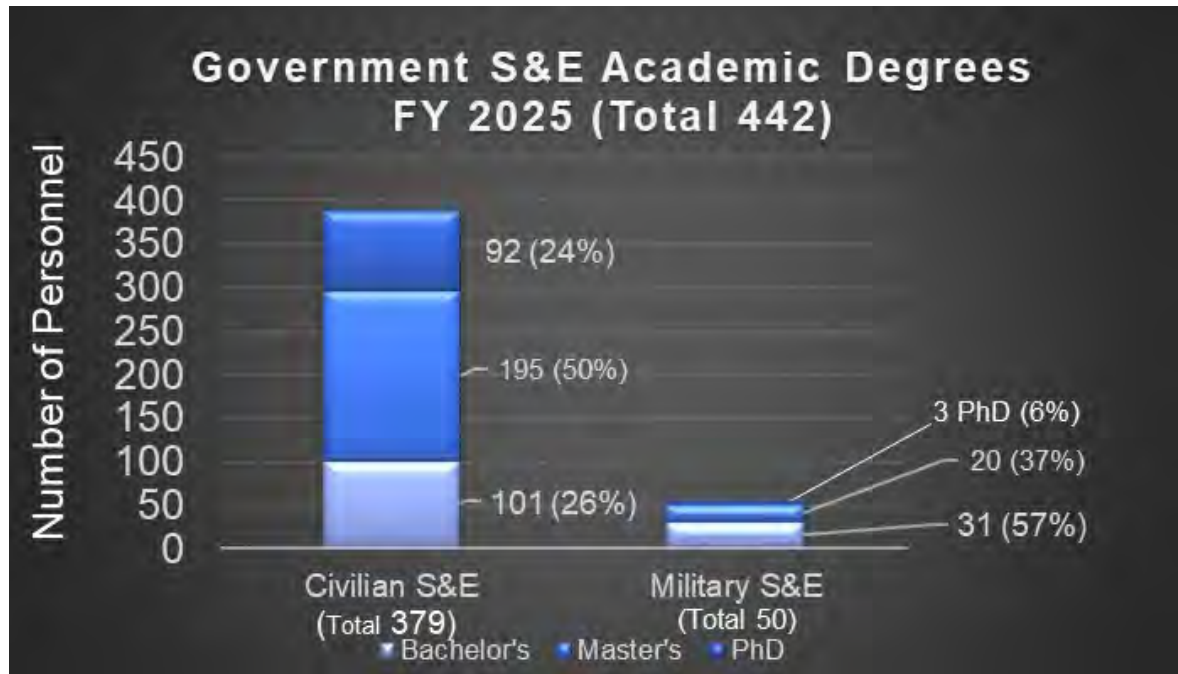


Source: <webeis> <RRS locator>
AFRL/RIB
as of 31Dec2024





Academic Talent



 **DEGREES**

120+

different disciplines
in multiple fields

230+

colleges/universities
attended

Workforce from

40

different states &
countries





Air Force Research Laboratory Information Directorate

QUICK REFERENCE TABLES FY24

Table 1: Personnel

Classification	Total	Multiplier	Impact Area Total
Appropriated Fund Military*	68	100.0%	68
Appropriated Fund Civilians*	775	92.9%	720
Local Contractors**	487	92.9%	452
Total Personnel	1330		1240

Table 2: Annual Payroll

Classification	Total	Multiplier	Impact Area Total
Appropriated Fund Military	\$11,287,088	100.0%	\$11,287,088
Appropriated Fund Civilians	\$118,370,966	92.9%	\$109,970,446
Local Contractors	\$78,342,880	92.9%	\$72,783,063
Total Annual Payroll	\$208,000,935		\$194,040,597

Table 1: * source: WebEIS, 30Sep2024

** source: AFRL/RI Comm-Computers System, 30Sep2024

Table 2: source: WebEIS, 30Sep2024

The Secretary of the Air Force for Financial Management (SAF/FM) specified the methodology for compiling the economic impact of an Air Force installation. This methodology is consistent with the methodology of the Office of the Secretary of Defense (OSD) Base Realignment and Closure (BRAC) Commission. The economic impact area of the Information Directorate consists of the counties of Herkimer, Madison, Oneida, Onondaga, and Oswego. 100.0% of the military personnel and 92.9% of the civilian personnel and local contractors reside in these five counties.

Total annual payroll is a summation of total gross wages, payroll taxes and fringe benefits.

Table 3: Expenditures within the Impact Area

	Annual Expenditures
Materials, Equipment, and Supplies	\$3,809,383
Facility Modernization/Sustainment	\$4,052,522
Service Contracts ¹	\$5,738,450
Research and Development ²	\$301,592,633
Education ³	\$129,693
Travel ⁴	\$3,481,314
Total Annual Expenditures	\$318,803,995

- 1 Includes only contracts in the economic impact area or contracts requiring the use of locally supplied goods and services.
- 2 Includes only Research and Development contracts granted to contractors in the economic impact area for scientific and technical work not elsewhere included.
- 3 Includes cost of registered classes in the economic impact area.
- 4 Includes travel expenditures of military and civilian personnel on temporary duty at the Information Directorate and local travel expenditures of AFRL/RI personnel on travel.

Source:

- WebEIS, 30Sep2024, for Facility Modernization/Sustainment, Service Contracts, Research and Development, and Materials, Equipment, and Supplies.

- AFRL/RIOW for Education

- AFRL/RIC and Defense Travel System for Travel

Table 5: Total Annual Economic Impact Estimate FY 2024

	Total Impact
Total Annual Payroll (Table 2)	\$194,040,597
Total Annual Expenditures (Table 3)	\$318,803,995
Estimated Annual Dollar Value of Jobs Created (Table 4)	\$87,119,703
Grand Total Annual Economic Impact Estimate for Five-County Impact Area	\$599,964,296

Approved for Public Release, Distribution Unlimited, AFRL-2025-2084

Table 4: Estimate of Number and Dollar Value of Indirect Jobs Created

The number of indirect jobs created is the mathematical product of the actual number of Information Directorate jobs (Military, Civilians, and Local Contractors) and the DoD Indirect Job Multipliers for the respective economic area. The estimated annual dollar value of jobs created is the mathematical product of the number of indirect jobs created and the average annual pay in the economic area (as published by the Bureau of Labor Statistics, US Department of Labor).

Classification	Impacted Jobs (from Table 1)	Multiplier	Indirect Jobs Created
Appropriated Fund Military	68	0.35	24
Appropriated Fund Civilians	720	1.21	871
Local Contractors	452	1.21	547
Total	1,240		1,442

Estimated Number of Indirect Jobs Created as a Result of the Information Directorate 1,442

Average Annual Pay for the Local Community² \$60,397

Estimated Annual Dollar Value of Jobs Created \$87,119,703

1 Economic Impact Analysis (EIA), Indirect Job Multipliers for the Air Force Installations

2 Bureau of Labor Statistics <http://www.bls.gov/>

Total Jobs Impacted

Classification	Impacted Jobs	Dollar Value of Impacted Jobs
Appropriated Fund Military	68	\$11,287,088
Appropriated Fund Civilians	720	\$109,970,446
Local Contractors	452	\$76,856,962
Total Direct Jobs	1,240	\$198,114,497

Indirect Jobs Created as a Result of the Information Directorate 1,442 \$87,119,703

Total Jobs Impacted Direct & Indirect 2683

Total Annual Dollar value of Impacted Jobs Direct & Indirect \$285,234,200

NOTE: Direct jobs are the number of Military, Civilians, and Local Contractors working at the Information Directorate in the Economic Impact Area in FY24.

Indirect jobs are the estimated number of jobs created as a result of the Information Directorate.



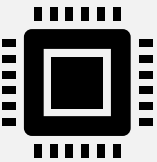
C2 Mission Area

Mission Statement: synchronize actions & accelerate decision making both at pace and scale to overwhelm our adversary



Communications Mission Area

Mission Statement: connect the force via a seamless, multi-domain, network of networks communications fabric across the enterprise



Cyber Superiority Mission Area

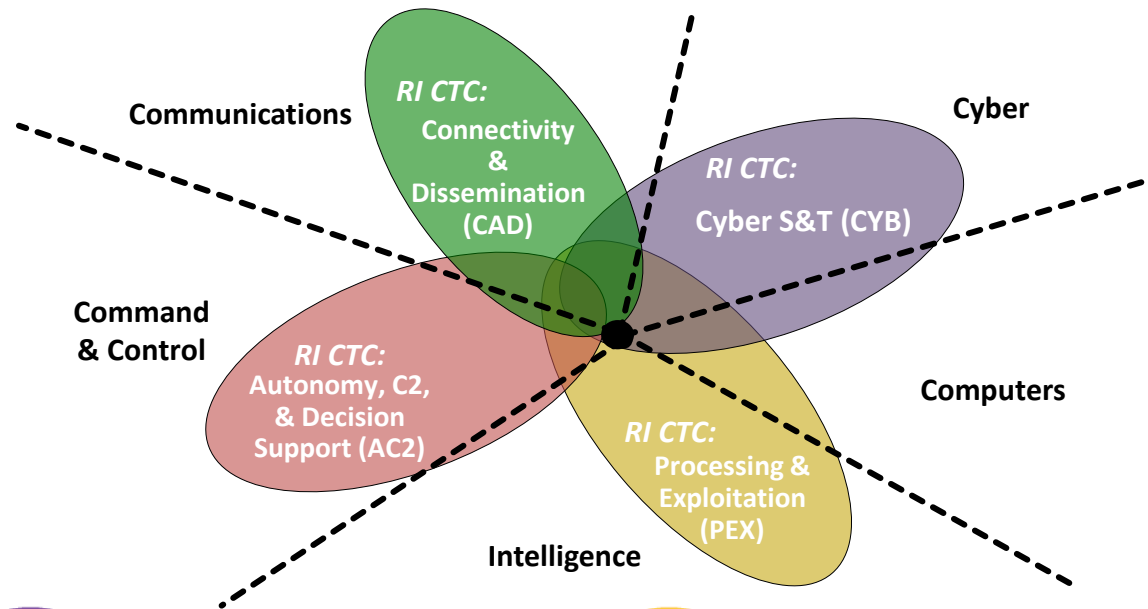
Mission Statement: lead the development of cyberspace science and technology necessary to ensure cyberspace superiority and support the conduct of full-spectrum cyberspace operations integrated with other domains.



Global Integrated ISR Mission Area

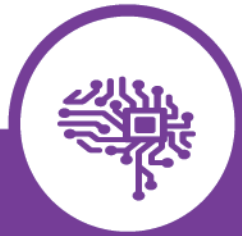
Mission Statement: collect and process decision-quality intelligence and act on it faster than adversaries can react. Develop and field sensing and sense-making capabilities to operate in contested environments, detect hard targets, and disseminate data.

Core Technical Competencies at RI



PUTTING THE RIGHT INFORMATION INTO THE RIGHT HANDS AT THE RIGHT TIME

Vision: Seamless, resilient networked communications fabric across the command, and control intelligence, surveillance and reconnaissance (C2ISR) enterprise



LEVERAGING AND SHAPING THE CYBER DOMAIN TO THE NATION'S ADVANTAGE

Vision: An Air Force equipped with technologies that enable our freedom to operate in cyberspace while denying the adversary the same.



EXPLOITING COMPUTING & ALGORITHMS TO TRANSFORM BIG DATA INTO INFORMATION

Vision: Innovator of technologies that process and exploit data in near real time, analyze massive collections over time and employ continuous learning to deliver asymmetric decision speed to the Air Force and Intelligence Community.

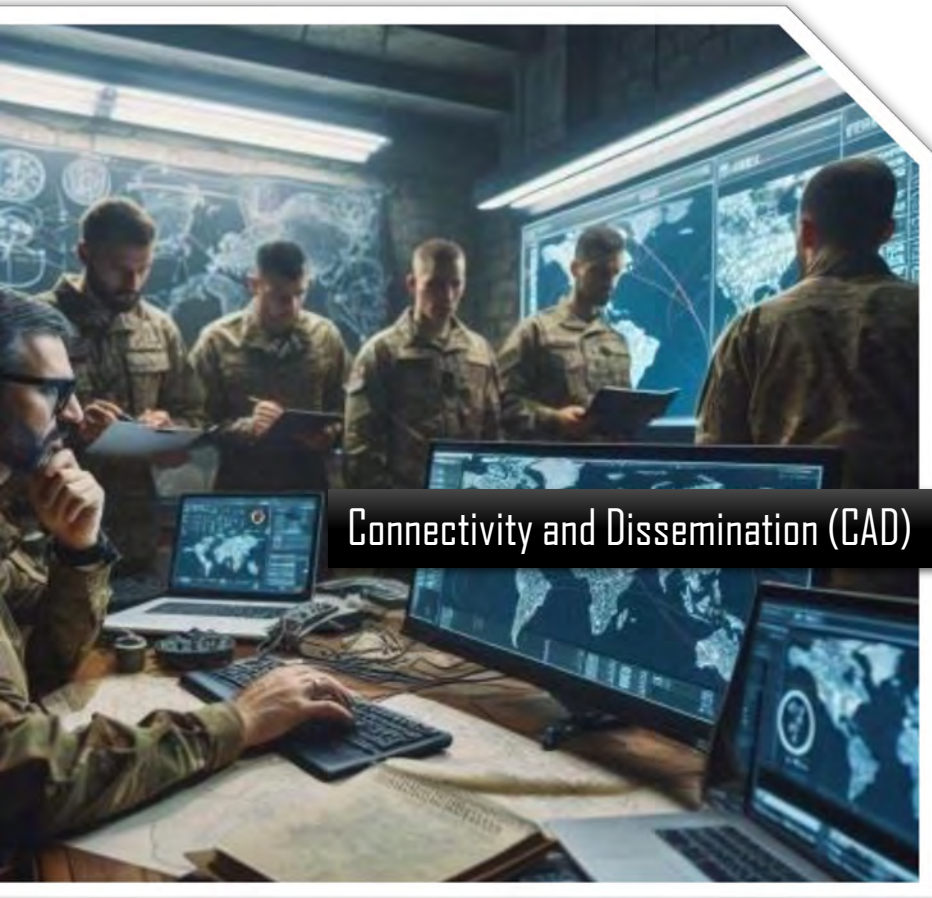


MASTERING COMPLEXITY OF MULTI-DOMAIN COMMAND & CONTROL

Vision: Mastering and imposing complexity to C2 joint all-domain operations in an evolving battlespace at speed and scale



Connectivity and Dissemination (CAD)



Vision

Seamless, multi-domain, *network of networks* connectivity fabric across the command and control intelligence, surveillance and reconnaissance (C2ISR) enterprise, assuring delivery of timely, secure, and actionable information to warfighters and systems.

Mission

Provide agile and secure mission-responsive communications and information sharing globally.

Goals

- Agile and secure communications and networks
- Platform agnostic connectivity
- Autonomous link discovery, creation and utilization
- Dissemination of information at need, securely

Putting The Right Information Into The Right Hands At The Right Time



Cyber Science and Technology (CYB)



Vision

An Air Force equipped with technologies that enable our freedom to operate in cyberspace while denying the adversary the same.

Mission

Deliver the science and technology necessary to ensure cyberspace superiority and support the conduct of full-spectrum, multi-domain, integrated cyber operations.

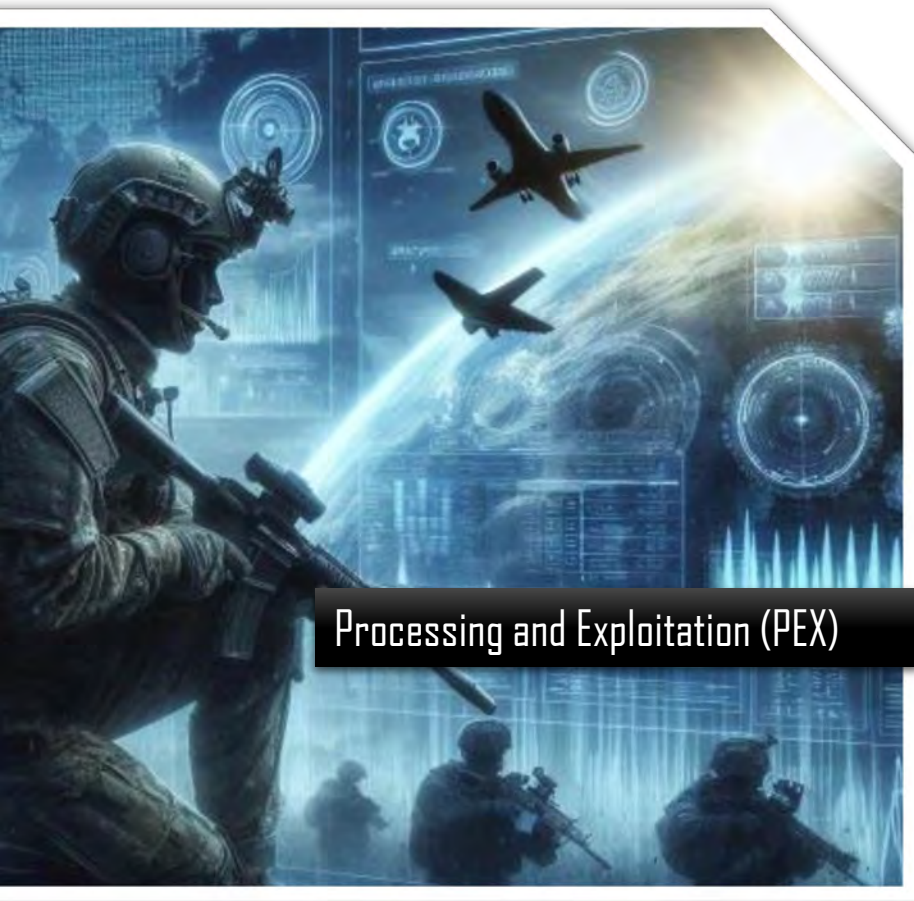
Goals

- Secure, composable, risk-based compute options
- Cyber operations integrated and on par with air & space
- Ability to conduct cyber operations agnostic to medium and geography

Leveraging And Shaping The Cyber Domain To The Nation's Advantage



Processing and Exploitation (PEX)



Vision

Innovator of technologies that process and exploit data in near real time, analyze massive collections over time and employ continuous learning to deliver asymmetric decision speed to the Air Force and Intelligence Community.

Mission

Deliver fast sense-making for situational awareness and adversarial insight for the AF, DoD, and Intelligence Community.

Goals

- Multi-INT correlation and fusion of massive amounts of intelligence, surveillance, and reconnaissance (ISR) and publicly available data
- Exploit targets in denied areas
- Adversarial and secure machine learning
- Dynamic, hybrid computing advancing neuromorphic, nanotech, and quantum systems to efficiently process ISR information

Exploiting Computing And Algorithms To Transform Big Data Into Information



Autonomy, Command & Control and Decision Support (AC2)



Autonomy, Command & Control and
Decision Support (AC2)

Vision

Mastering and imposing complexity to command & control future multi-domain operations in an evolving battlespace with speed and scale.

Mission

Deliver revolutionary, trusted, affordable information technologies for agile, resilient and distributed Air Force command & control and autonomous systems.

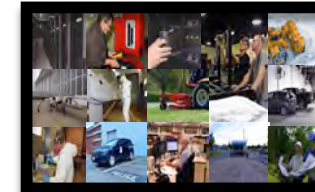
Goals

- Master complexity through development of adaptive command & control systems-of-systems and services
- Control, impose and synchronize complex multi-domain effects chains
- Harness machine intelligence to increase command & control speed and scale of operations
- Realize large-scale multi-agent systems for autonomous planning, tasking and execution

Mastering Complexity of Multi-domain Command & Control



AFRL/RI Location Info



- ROME, NEW YORK
- 65 Acre Campus
- 30 Laboratories & Facilities
- 882,000 Sq Ft Floor Space
- Offsite specialty locations in Newport, NY and Stockbridge, NY



AFRL/RI Labs, sites and facilities



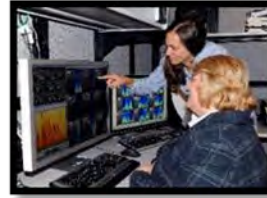
Machine Intelligence
for ISR Laboratory



Situation Awareness
Laboratory



Cyber Experimentation
Environment (CEE)



Audio Processing Lab



Operational Information
Management Lab



Integrated Intelligence
Innovation Facility (I3F)



Newport Remote Research Site



Secure Embedded High
Performance Computing



Small Unmanned Aerial System
Experimental Capability (SUAS-EC)



Command and Control
Technology Center (C2TC)



High Performance Computing Facility



Advanced Computing
Applications Laboratory



Quantum Information
Science Facility



Quantum Communications
Laboratory



Nanotechnology & Computational
Intelligence Laboratory



Corporate Collateral
Facility (CCF)



Cyber Integration &
Transition Environment



K5 Laboratory



Corporate Research and
Development Server
Facility (CRDSF)



Microwave and
Optical
Communication
Range



RF Technology Center



Cyber Operations
Technology
Facility (COTF)



Network-Centric Integration &
Interoperability Facility (NCIIF)



Command and Control
Concept Center (C2CC)



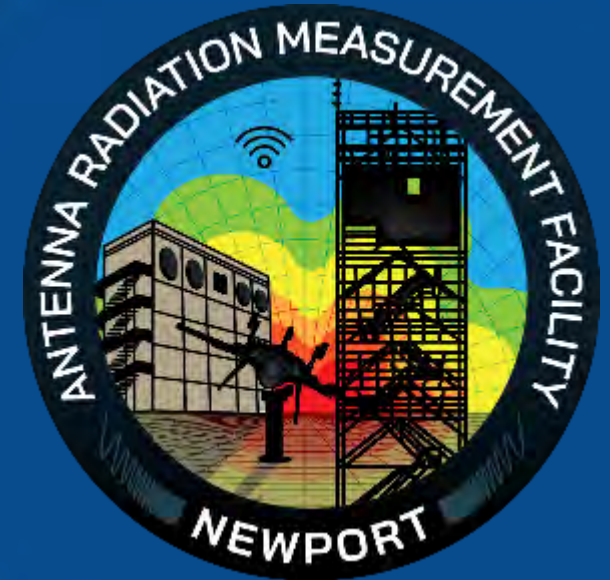
Stockbridge Remote Research Site



AFRL/RI Newport Site

Far Field, Elevated Outdoor Antenna Test Range

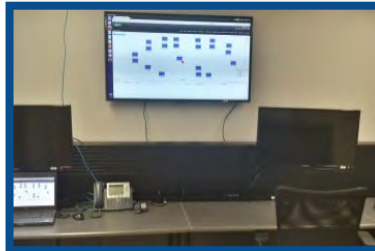
- 78 Acres
- 360° pattern measurement
- Established in 1972
- Ideal geography
- Essential measurements of the F-35 aircraft antenna patterns
- Inflatable reflector antennas for SOCOM
- 12 Commercial Test Agreements
- Aircraft and vehicle antenna performance measurements
- Critical capability for future aircraft/vehicle design and development
- Terahertz comm demonstration - provides LPI/LPD/AJ air-to-air comm links



AFRL/RI Stockbridge Site

RF and Small UAS Experimental Facility

- 300 acre flexible test site, varying in relative distance, topology and foliage density
- Heavy-duty turntable with A 200' high arched measurement probe – large aircraft and vehicle capable
- 120' walkup tower for LOS and optical links
- Controllable contested environment
- All weather, full season, configurable RF capability
- C4ISR, cyber, spectrum, networking
- Flexible frequency authorizations
- SUAS airfield
- Fixed wing and VTOL platforms
- Trained flight personnel
- Experiment, management and control facility
- Flexible laboratory space
- Operations and control room

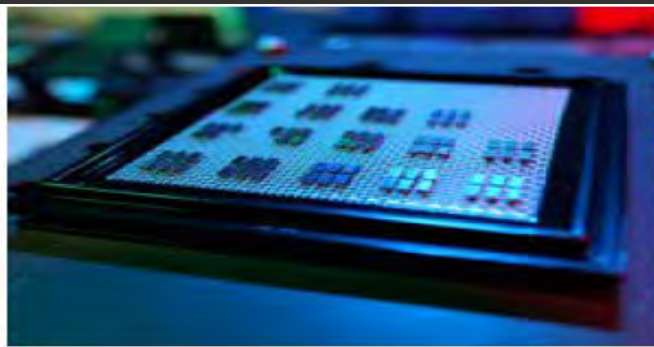




AFRL/RI Extreme Computing Facility

A Computing Challenge Space

- Foundational advances in computing architectures
- Quantum
- Neuromorphic
- Nanoelectronic
- Machine Learning
- Artificial Intelligence

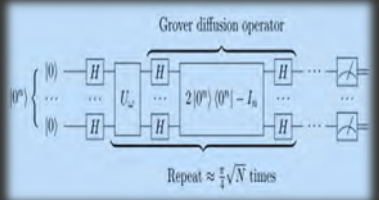


Quantum Research at the AFRL Information Directorate (RI)



Quantum Node & Interconnect Development

Trapped Ions, Integrated Quantum Photonics & Superconducting Qubits
Heterogeneous qubit interfaces for entanglement distribution applications
Verify and validate quantum networking components – classical or quantum on heterogeneous network



Quantum Computing Readiness

In-house algorithm development to assess viability of solving DAF problems
Government led honest broker evaluation of the claims and progress of industry
Commercial system engagement through cloud access to understand how to use quantum systems



Fieldability & Ruggedization

Qubit resilience for deployment
Test at our local field sites
Operate a low SWAP mobile quantum node



Objective: Demonstrate a multi-domain quantum network to exploit disruptive quantum technologies for DoD advantage



AFRL/RI Neuromorphic Computing

Brain-inspired, extremely low SWaP, intelligent computing at the-edge in dynamic & contested mission environments

- Neurosynaptic processors
- Nanoelectronics

AGILE CONDOR

- Real time situational awareness
- Neuromorphic architecture on-board
- Actionable intelligence with anomaly detection models, target recognition, and data fusion





Innovare Advancement Center

World-Class Facility for a Global Network of Researchers

Agility + Innovation + Partnerships

- Led by (AFRL/RI) and Griffiss Institute
- Driving Critical Innovations Accelerate Next-Gen Tech
- Advancing Artificial Intelligence/Machine Learning, Cybersecurity, and Quantum Computing
- Inspiring the Next Generation, Elevating New Talent for the Future and Beyond
- 150,000 square feet
- 13,000 square feet of Open Research Area
- Located in the Heart of New York State
- Two Leading-Edge Quantum Labs
- Event Space: Training Areas, Conference, & Breakout Rooms
- Co-Located Small Unmanned Aircraft Systems (sUAS) Site
- Two Neuromorphic/Nanoelectronics-Focused Labs





Innovare Advancement Center | *Agility + Innovation + Partnerships*

INNOVARE WILL BOLDLY...

- Build Rome's runway to the world, engaging a global community of **100 partners** to introduce game-changing capabilities built in Air Force core strengths in AI/ML, cyber and quantum.
- Advance the economy with **100 entrepreneurial ventures** and tech startups.
- **Elevate by 10%, our community's intellectual leadership** in AI/ML, cyber and quantum

INNOVARE ASPIRE TECHNICAL CHALLENGES

- Neurosymbolic C2
- AI-Enabled Change Detection Non-Traditional Events
- Internet of Things (IoT) Living Laboratory
- Harnessing Weird Machines
- Multi-Source Workflow for Event Detection and Evaluation

INNOVARE OPPORTUNITIES



AFRL



The State University
of New York



GRIFFISS
INSTITUTE

BROOKHAVEN
NATIONAL LABORATORY

Google IBM



AF





Technology Transfer



Created to ensure Air Force S&E activities are transferred or shared with state and local governments, academia and industry.

The exchange of knowledge, expertise, equipment, and testing facilities leverages DoD research and development investment

FY24 OPEN Agreements per AF Org per Transfer Mechanism			
AF Org	CRADA	EPA	CTA
711 HPW	2	2	
711 HPW/RH	39	5	2
AFRL/SB	128	16	1
RD	5	41	
RI	93	116	8
RQ	39	36	2
RV	14	3	1
RW	24	5	
RX	71	24	
RY	27	9	



172

Active Licenses
For FY24

FY24 OPEN Agreements
per AF Org per
Transfer Mechanism



93 CRADA's
116 EPA's
8 CTA's



Information Directorate Engagements



INDUSTRY ENGAGEMENTS



ACADEMIC ENGAGEMENTS

Academic Institutions, Partnerships, EPAs, and Visiting Faculty Research Program





Academic Partnerships | Educational Partnership, CRADA & Visiting Researchers

 Air Force Academy	Johns Hopkins University	 Purdue University	Cities	York College of Pennsylvania
Arizona State University	Kansas State University	Rose-Hulman Institute of Technology	University of Missouri, Kansas City	• Central State University
Auburn University	LaSalle University	Rutgers – State University of New Jersey	University of Nevada	• Dillard University
Augusta University	Louisiana State University	Stevens Institute of Technology	University of Notre Dame	• Florida A&M University
Boise State University	Louisiana Tech University	Temple University	University of Oklahoma	• Howard University
Brescia University	Massachusetts Institute of Technology	 Toyota Technological Institute at Chicago (TTIC)	University of Pennsylvania	• Monash University
Brown University	Michigan State University	Universidad Ana G. Mendez	University of Puerto Rico	• Norfolk State University
Carnegie Mellon University	Michigan Technological University	University of Arkansas	University of Southern Alabama	• North Carolina Agricultural
Clayton State University	Minnesota State University	University of Colorado, Denver	University of South Carolina	• & Tech State University
Colorado State University	Missouri University of Science & Technology	University of Connecticut	University of Southern Mississippi	• Prairie View A&M
 Cornell University	Montana State University	University of Illinois	University of Tennessee	• Tennessee State University
Dartmouth College	New Jersey Institute of Technology	University of Kansas	University of Tulsa	• Texas Southern University
 Duke University	Northeastern University	University of Maryland	Utah State University	• Tuskegee University
Fairleigh Dickinson University	Northern Arizona University	University of Massachusetts, Amherst	Vanderbilt University	Navajo Technical University
Georgia Tech	Northwestern University	University of Massachusetts, Dartmouth	Villanova University	
Hamilton College	Norwich University	University of Michigan	Washington University in St. Louis	
Harvard University	Oklahoma State University	University of Minnesota, Twin	Western Michigan University	
Imperial College London	Pennsylvania State University		Wichita State University	
Indiana University of Pennsylvania	Princeton University		Worcester Polytechnic Institute	
Iowa State University				



Denotes AFRL Regional Hubs



Denotes Centers of Excellence



AIR & SPACE
STEM Outreach
A Force to Shape the Future

K-12 STEM OUTREACH

The Air Force Research Laboratory Information Directorate K-12 STEM Outreach Program offers a variety of programs and services that effectively engage, inspire and attract the next generation of STEM talent.

**PARTNERSHIP INTERMEDIARY AGREEMENT
WITH THE GRIFFISS INSTITUTE TO SUPPORT
ENTIRE AIR FORCE STEM PROGRAM.**

STEM SUMMER CAMPS

- LEGO Robotics Camp
- Cyber Summer Camp
- Arduino Camp
- Engineering Camp
- Drone Camp
- Quantum Camp
- Sphero Camp

ORGANIC PROGRAMS

- Annual Challenge Competition
- Take Your Student to Work Day
- Lab & Innovare Tours

SUPPORTED NATIONAL PROGRAMS

- FIRST LEGO League
- FIRST Tech Challenge
- FIRST Robotics Competition
- CyberPatriot

ADDITIONAL SUPPORTED PROGRAMS

- Drone Soccer
- ABCs of STEM
- BOCES Internships
- Science & Career Fairs

STUDENTS IMPACTED

EVENTS SUPPORTED

DoD S&E's INVOLVED

HOURS OF STEM ACTIVITIES

70+

100+

3000+

1,450+





INFORMATION DIRECTORATE

Global Persistent Awareness
Resilient Information Sharing
Rapid, Effective Decision-Making
Complexity, Unpredictability, and Mass
Speed and Reach of Disruption and Lethality

Building a more lethal force – modernizing key capabilities
Strengthening our alliances and attracting new partners
Reforming our organization for greater performance and affordability

- Innovating at speed
- Employing rapid, iterative approaches for development → fielding

**WE UNDERSTAND THE URGENT
NEED TO “OUT-THINK, OUT-
MANEUVER, OUT-PARTNER,
AND OUT-INNOVATE...”**



INTERACTIVE SESSION

ACADEMIA

- Grants
- Partnerships

INTERNAL

- Department of the Air Force Challenge
- AFWERX Spark Program
- AFRL CC's Challenge

SMALL BUSINESS

- Open Innovation Challenges
- Tech Accelerators
- AFRL's Innovation Institutions
- IP Licensing
- Small Business Innovation Research (SBIR)



For more information visit

[AFRESEARCHLAB.COM](https://afresearchlab.com)



SCAN ME

INDUSTRY

- AFRL Institutes
- AFWERX, SpaceWERX
- AFVentures
- Open Solicitations
- beta.sam.gov
- Defense Innovation Marketplace

Partnering with AFRL